Signal-Driven Derived Data

post_save / post_delete receivers keep aggregates and indicator chains consistent; update() avoids recursion

ADMINISTRATIVE/SIGNALS.PY

Neighborhood
saved / deleted

_recompute_population()

District
currentPopulation · density · urban_area

update()

City
currentPopulation · density · urban_area

update()

Province
currentPopulation · density · urban_area

Sums both population and urban_area per level.

re-enter cascade

PHYSICALENV/SIGNALS.PY

Building created (map editor, id < 0)
building_created · DAG New_Units → LandCover

urbanize_around(): 10 m buffer → urban fabric
cut out of the city's latest-year land cover (ST_Difference), inserted as one polygon · landcover_rewriting / _created → mainMap journal, undone by Restore

LandCoverVector saved
schedule_urban_area_recompute(city_id)

One set-based UPDATE … RETURNING
ST_Intersection with urban-fabric polygons (latest year), skipped when fully inside; ST_Area in storage-CRS metres

CORE/SIGNALS.PY

RASTER post_save
export_raster_to_cog()

VECTOR save/delete
bump_layer_version() → GeoJSON cache

WATERSUPPLY/SIGNALS.PY + SAVE()

Population (cascade)

ExtractionWater

ConsumptionCapita

ImportedWater

TotalWaterDemand

TotalWaterProduction

SupplySecurity

Service_Time

OPEX
extraction · imported
treatment · pipes

WaterTreatment · MeteredResidential · PipeNetwork → OPEX

NonRevenueWater
→ same-year ILI

Coverage
PipeNetwork + UsersLocation

BULK WRITES · IMPORTER/BATCHING.PY::DEFERRED_CASCADES()

= administrative.deferred_population_cascade() + physicalEnv.deferred_urban_area()
+ watersupply.deferred_water_refresh()
Inside the block, receivers only record the affected parent; on exit each cascade runs once per parent (errors during that flush are logged, never raised over the original exception).
Always call schedule_urban_area_recompute() rather than the recompute itself, so an enclosing deferral is respected.